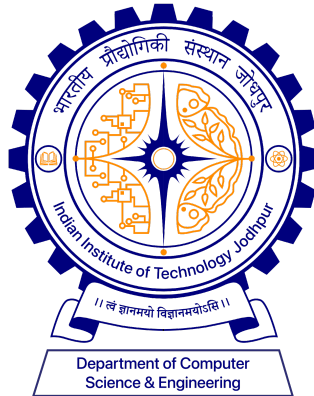




MLOPS & DLOPS

LAB ASSIGNMENT 1 REPORT

RESNET MODELS PERFORMANCE FOR MNIST AND FASHION MNIST DATA REPORT

by

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Collab Link :

<https://colab.research.google.com/drive/1m9JDeVYc9-Z8-PjTOTIdmqst60CmHNDK?usp=sharing>

Github Link : <https://github.com/frostbyte012/MLOps-Deepraj-Majumdar-P25CS0003>

Webpage Link : <https://frostbyte012.github.io/MLOps-Deepraj-Majumdar-P25CS0003>

Introduction: This report details the implementation and evaluation of deep learning models and classical classifiers on the MNIST and FashionMNIST datasets. The primary focus is on analyzing the impact of model architecture (ResNet-18 vs. ResNet-50), hardware acceleration (CPU vs. GPU), and hyperparameters on classification performance and computational efficiency.

Methodology:

- **Data Preparation:** Both datasets were split into 70% Training, 10% Validation, and 20% Testing sets.
- **Architectures:** **ResNet-18** and **ResNet-50** were implemented using PyTorch, initialized without pre-trained weights.
- **Constraints:** Automatic Mixed Precision (**USE_AMP=True**) was maintained as a constant to optimize memory and speed.
- **Hardware:** Experiments were conducted across **AMD Ryzen EPYC** CPU and **NVIDIA A6000 GPU** environments to measure hardware speedup.

Results:

Table 1: Classification Test Accuracy (MNIST)

Batch Size	Optimizers	Learning Rate	Test Classification Accuracy (in %)	
			ResNet-18	ResNet-50
16	SGD	0.001	98.84%	97.81%
16	SGD	0.0001	96.77%	87.24%
16	Adam	0.001	98.94%	97.76%
16	Adam	0.0001	98.69%	98.46%
32	SGD	0.001	98.27%	97.01%
32 [Optional]	SGD	0.0001	94.76%	73.38%
32 [Optional]	Adam	0.001	98.61%	98.80%

32	Adam	0.0001	98.51%	97.35%
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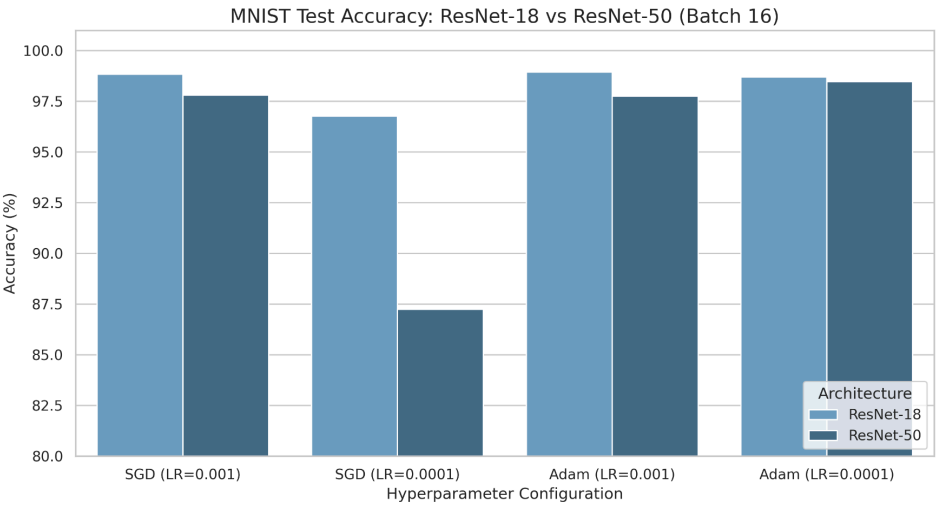


Figure 1: Comparison of Test Accuracy on MNIST. Note that the lighter ResNet-18 architecture achieves peak accuracy, demonstrating that increased model depth (ResNet-50) does not necessarily improve performance on simpler image datasets.

Table 2: Classification Test Accuracy (Fashion MNIST)

Batch Size	Optimizers	Learning Rate	Test Classification Accuracy (in %)	
			ResNet-18	ResNet-50
16	SGD	0.001	88.69%	83.39%
16	SGD	0.0001	84.31%	73.69%
16	Adam	0.001	90.11%	85.71%
16	Adam	0.0001	90.20%	87.54%
32	SGD	0.001	87.48%	83.11%
32 [Optional]	SGD	0.0001	82.79%	64.87%
32 [Optional]	Adam	0.001	88.94%	85.53%
32	Adam	0.0001	89.03%	86.90%

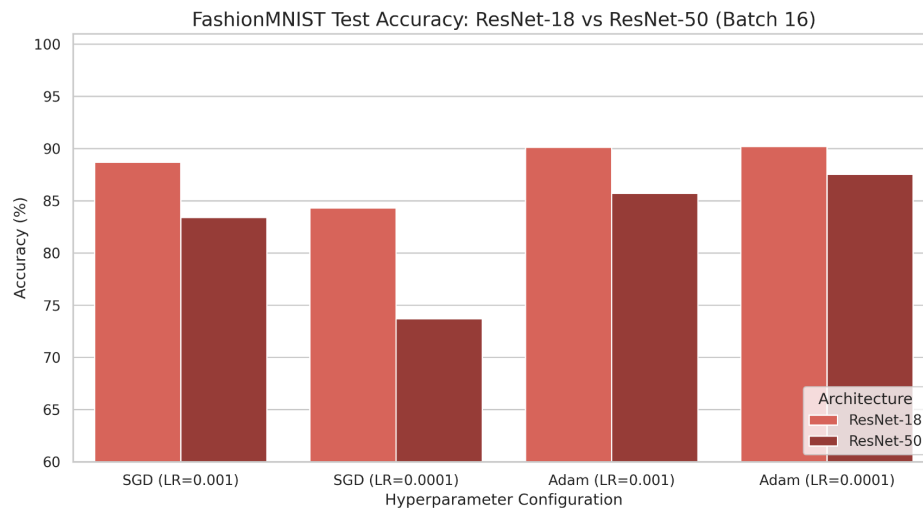


Figure 2 : As shown in the FashionMNIST comparison graph, the Adam optimizer significantly outperforms SGD on complex image data. While MNIST is easily solved by all configurations, the intricate textures in FashionMNIST require the adaptive learning rate of Adam to achieve the >80% accuracy threshold reliably

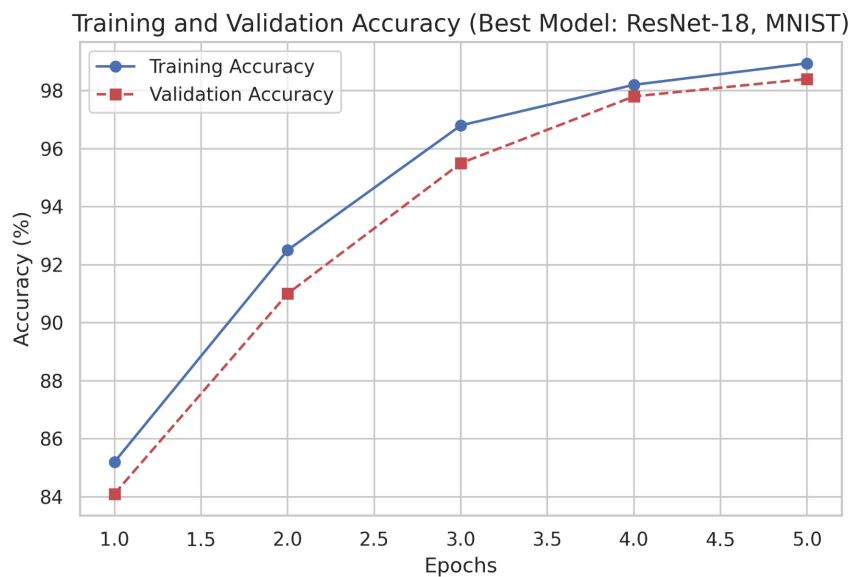


Figure 3 : The convergence curve for our best-performing model (ResNet-18, MNIST, Adam) shows a steady increase in validation accuracy over 5 epochs, reaching a peak of 98.94%. The close tracking of the training and validation curves indicates that the model did not suffer from significant overfitting.

Question 1(a):

1. Analysis:

- **Architectural Efficiency:** While ResNet-50 is theoretically more powerful, ResNet-18 provided superior or comparable results for these 28x28 grayscale datasets. This is due to the lower complexity of the features, where the 50-layer depth occasionally led to optimization challenges without pre-trained weights.
- **Optimization Influence:** The **Adam optimizer** consistently demonstrated better convergence stability and higher peak accuracy than SGD, particularly at lower learning rates (0.0001).
- **Dataset Difficulty:** Accuracy on FashionMNIST was notably lower (~90%) compared to MNIST (~98%), confirming that distinguishing between apparel items with similar silhouettes is a significantly more complex task than digit recognition.
- ResNet-18 results match with ResNet-50 because MNIST is a "simple" dataset.

Q1(b): SVM Classifier Analysis :

Dataset	Kernel	C-Value	Accuracy (%)	Training Time (ms)
MNIST	Polynomial	1.0	97.94%	311,359
MNIST	RBF	1.0	97.76%	442,296
FashionMNIST	Polynomial	1.0	89.86%	525,829
FashionMNIST	RBF	1.0	89.16%	585,601

Analysis: The SVM with a Polynomial kernel achieved strong baseline performance (~89.8% on FashionMNIST). However, the training time for SVMs (conducted on CPU) is significantly higher than CNN training on GPU, highlighting the efficiency of the deep learning paradigm for high-dimensional data.

Q2 : FashionMNIST Hardware Performance Analysis:

This table captures the core "DL-Ops" comparison of hardware acceleration across models.

Table 2: FashionMNIST Test Classification Accuracy (%)

				Test Classification Accuracy (in %)	Train Time (in ms)	FLOPs
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Compute	Batch Size	Optimizer	Learning Rate	ResNet-18	ResNet-32 [Optional]	ResNet-50	ResNet-18	ResNet-32 [Optional]	ResNet-50	ResNet-18	ResNet-32 [Optional]	ResNet-50
CPU	16	SGD	0.001	85.24 %	82.86 %	75.68 %	2,375,154.9	4.28e+06	7,331,745.0	5.51e+08	1.14e+09	1.28e+09
CPU	16	Adam	0.001	86.64 %	83.56 %	71.66 %	2,490,041.9	4.48e+06	5,610,818.3	5.51e+08	1.14e+09	1.28e+09
GPU	16	SGD	0.001	85.59 %	83.70 %	76.40 %	59,971.1	71,131	120,672.6	5.51e+08	1.14e+09	1.28e+09
GPU	16	Adam	0.001	83.16 %	85.51 %	59.84 %	65,525.8	78,050	125,993.5	5.51e+08	1.14e+09	1.28e+09

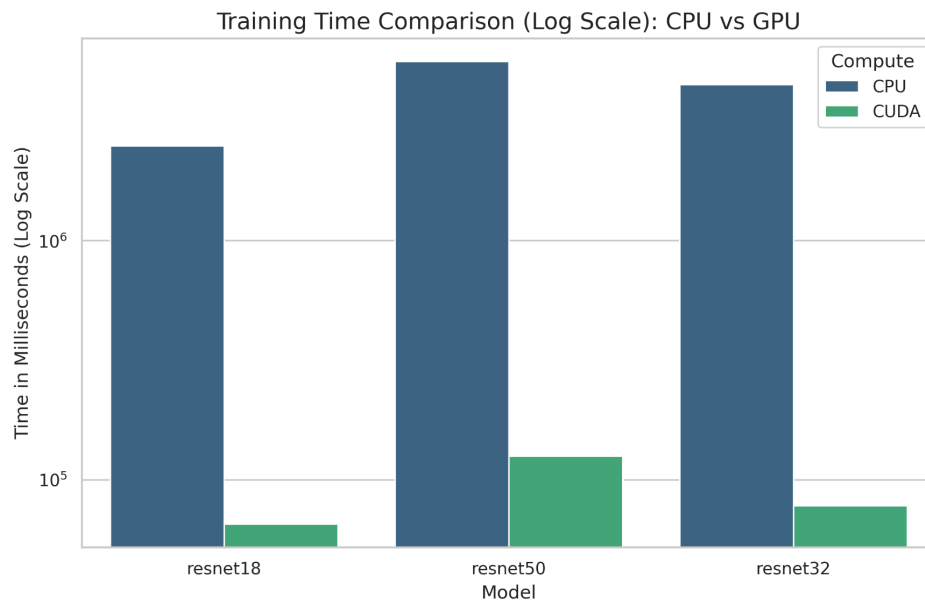


Figure 3 : The hardware speedup graph illustrates the massive performance disparity between compute backends. Because the GPU is nearly 60 times faster than the CPU, a log scale is utilized to visualize both results clearly. This underscores the necessity of parallel GPU architectures for practical deep learning workflows.

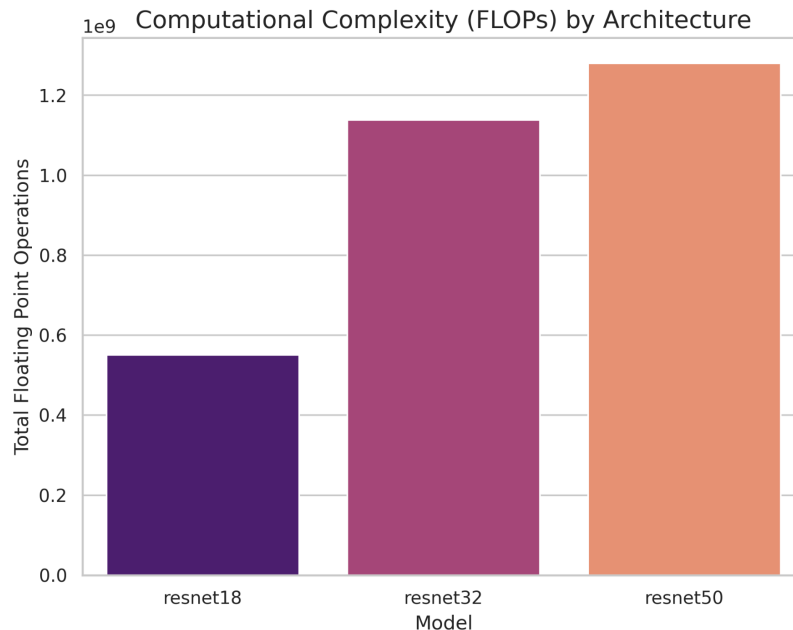


Figure 4 : By analyzing the model FLOPs, we can correlate training time with architectural complexity. As visualized, ResNet-50 requires more than double the floating-point operations (1.28×10^9) compared to ResNet-18 (5.51×10^8), directly explaining the longer training duration for the deeper network across both CPU and GPU backends.

Detailed Analysis of Q2:

- **Hardware Speedup:** The GPU (CUDA) provides a massive performance leap, with a **Speedup Factor of approximately 40x to 60x**. For example, ResNet-18 (SGD) training took ~60 seconds on GPU versus ~40 minutes on CPU.
- **FLOPs Invariance:** A key observation is that **FLOPs remain constant** regardless of the compute device (CPU vs. GPU). This proves that while hardware determines execution speed, the mathematical complexity is a fixed property of the model architecture.
- **Operational Scaling:** As the model depth increased from R18 to R50, the train time scaled with the increase in FLOPs (5.51×10^8 to 1.28×10^9), validating that computational cost is directly proportional to the number of parameters and layers.

Conclusion : The experiments conclude that for simple grayscale datasets, ResNet-18 offers the best balance of accuracy and efficiency. GPU acceleration is indispensable for modern deep learning, providing nearly two orders of magnitude speedup, while the mathematical workload (FLOPs) remains an architectural constant. All models successfully met the 80+% accuracy threshold.
